

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

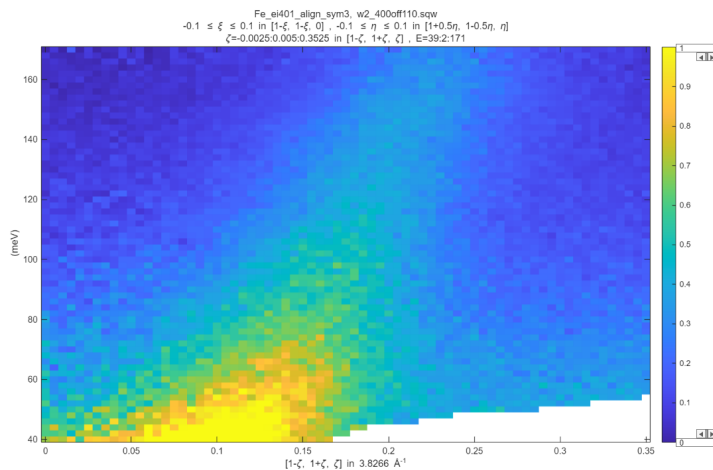
```

```

selectCuts= 'Ei400_off110GPmaxFF';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end

%mi = maps_instrument(200,600,'S');
mi = maps_instrument(401,600,'S');
%mi = maps_instrument(786,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GP";

dE_step = 2; %original energy transfer step data were binned to. No point in going finer
half_dE = 5; % half width of data binning

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```

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

```

```

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);

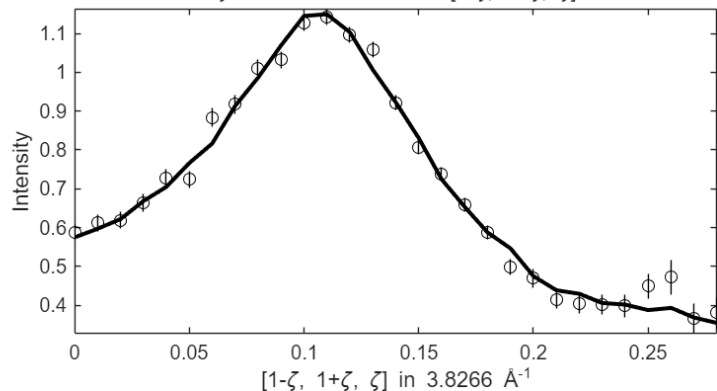
```

```

*****
**** En = 44±5 going to 194
*****

```

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.29\pm0.0074$; $J0=6.7\pm0.033$; $\gamma=4.23635\pm0.0854536$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $39 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

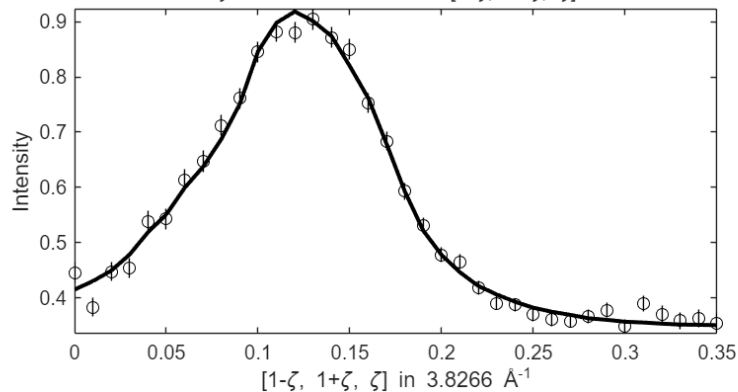


```

*****
**** En = 54±5 going to 194
*****

```

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.19\pm0.0031$; $J0=6\pm0.028$; $\gamma=3.12421\pm0.0383619$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

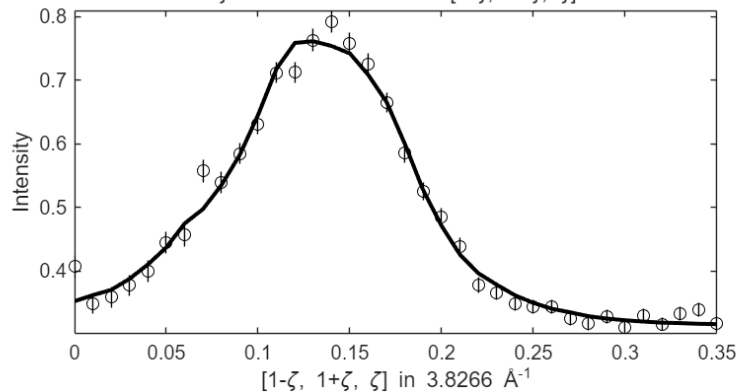


```

*****
**** En = 64±5 going to 194
*****

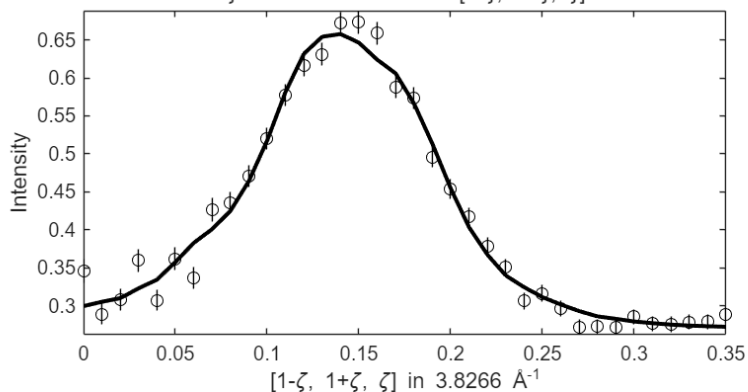
```

Fe_ei401_align_sym3, w2_400off110.sqw
 S=0.16±0.0032; J0= 5.7±0.034; gamma=3.82891±0.0613172
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



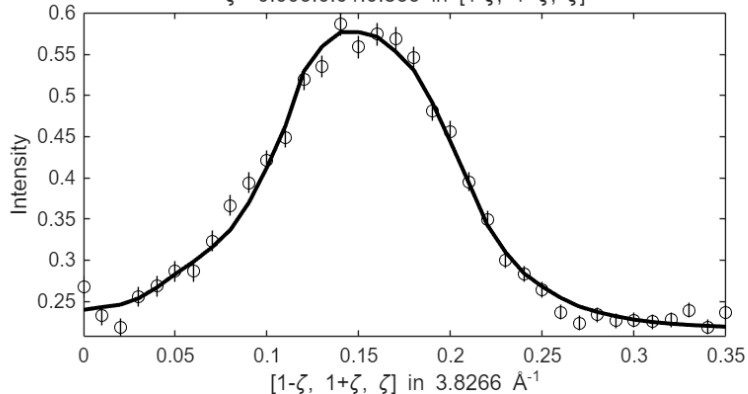
**** En = 74±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 S=0.15±0.0034; J0= 5.4±0.03; gamma=4.83296±0.0817231
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



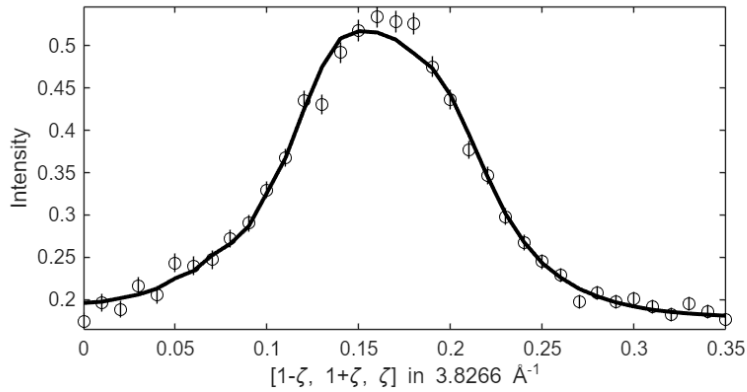
**** En = 84±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 S=0.15±0.0029; J0= 5.1±0.025; gamma=6.08127±0.0868703
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



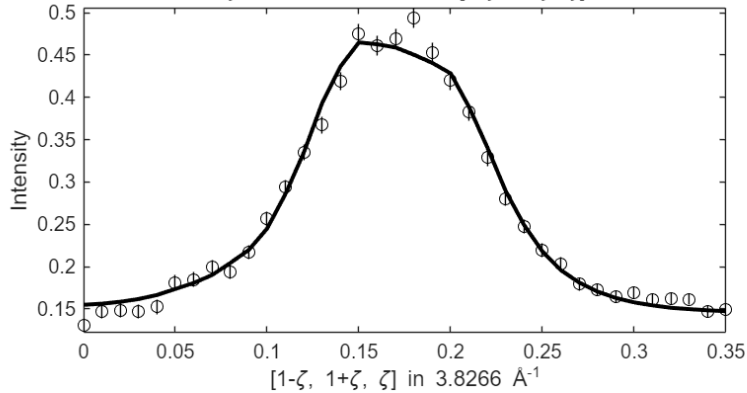
**** En = 94±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 S=0.14±0.0029; J0= 4.9±0.024; gamma=7.34394±0.117117
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



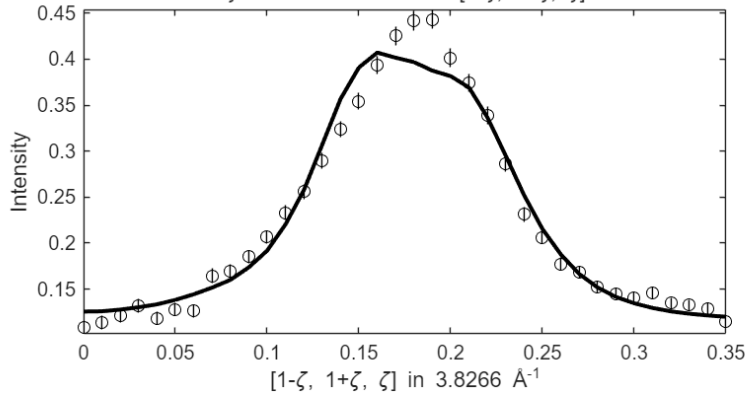
**** En = 104±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 S=0.13±0.0025; J0= 4.7±0.019; gamma=6.77434±0.103601
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



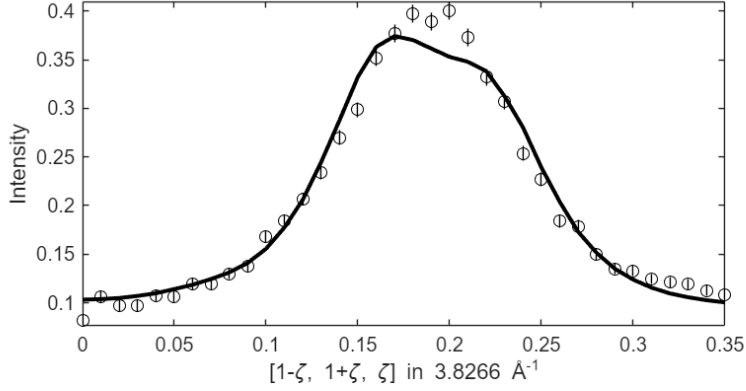
**** En = 114±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 S=0.13±0.0038; J0= 4.5±0.029; gamma=8.89213±0.228636
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



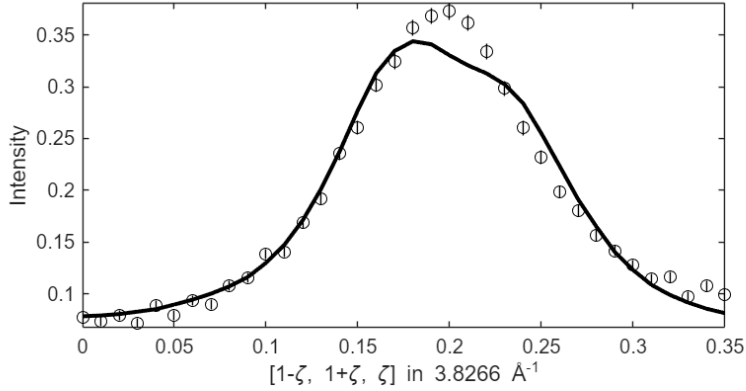
**** En = 124±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.14\pm0.0041$; $J0= 4.3\pm0.025$; $\gamma=12.3151\pm0.351237$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



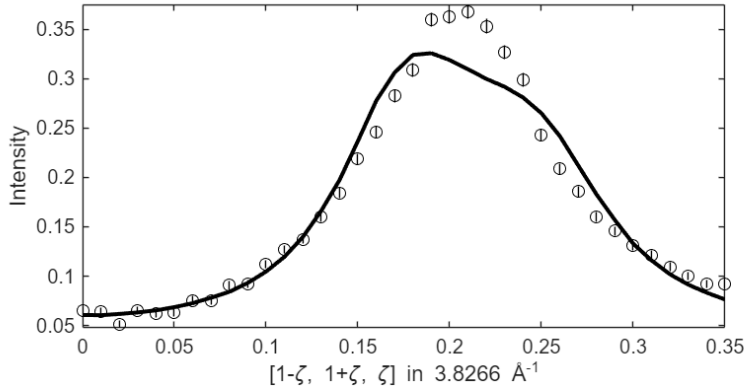
 **** En = 134±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.17\pm0.0082$; $J0= 4.1\pm0.024$; $\gamma=19.1441\pm1.33718$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



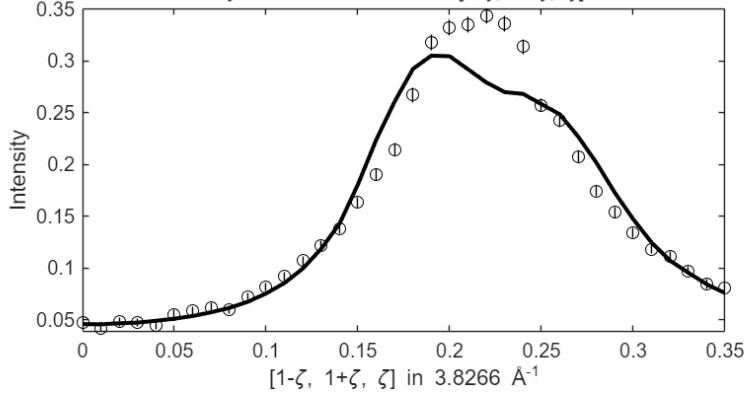
 **** En = 144±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.19\pm0.015$; $J0= 4\pm0.03$; $\gamma=22.952\pm3.38336$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



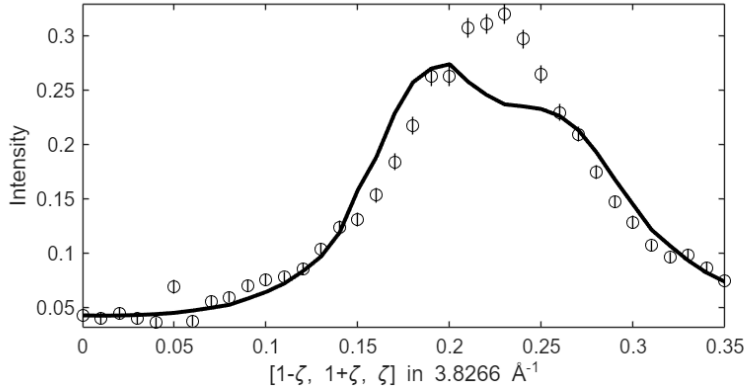
 **** En = 154±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.19\pm0.013$; $J_0= 3.8\pm0.029$; $\gamma=22.8158\pm3.14549$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



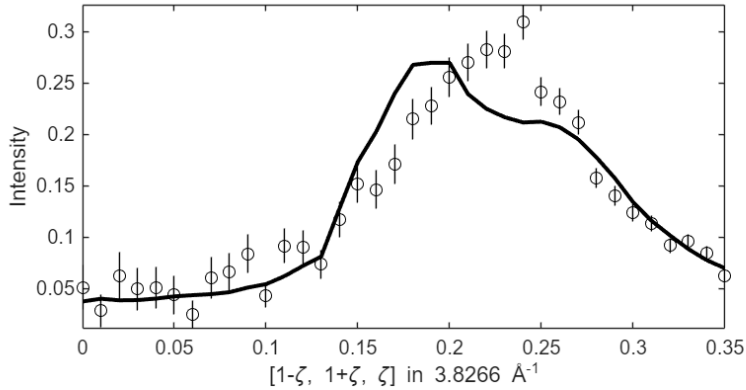
 **** En = 164±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.17\pm0.015$; $J_0= 3.7\pm0.034$; $\gamma=22.9081\pm3.72971$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



 **** En = 174±5 going to 194

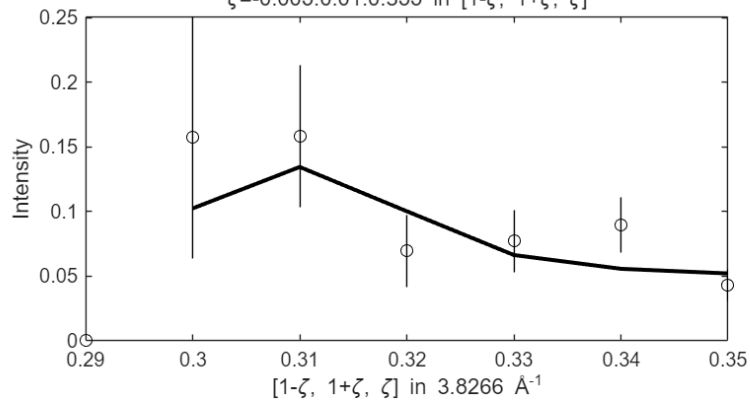
Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.16\pm0.017$; $J_0= 3.5\pm0.035$; $\gamma=20.8014\pm4.02517$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $169 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



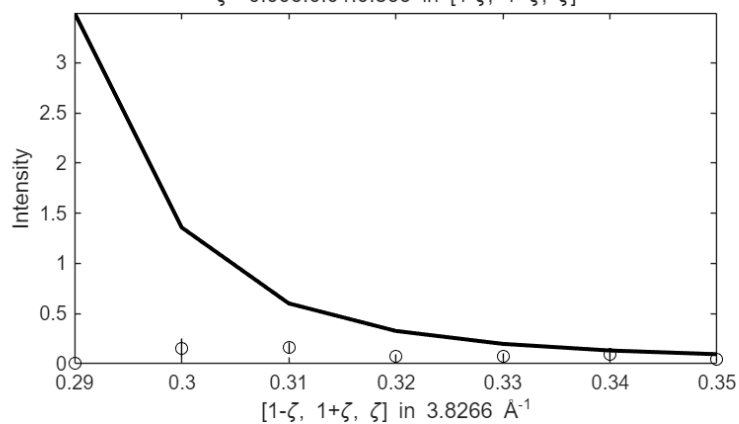
 **** En = 184±5 going to 194

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.7778

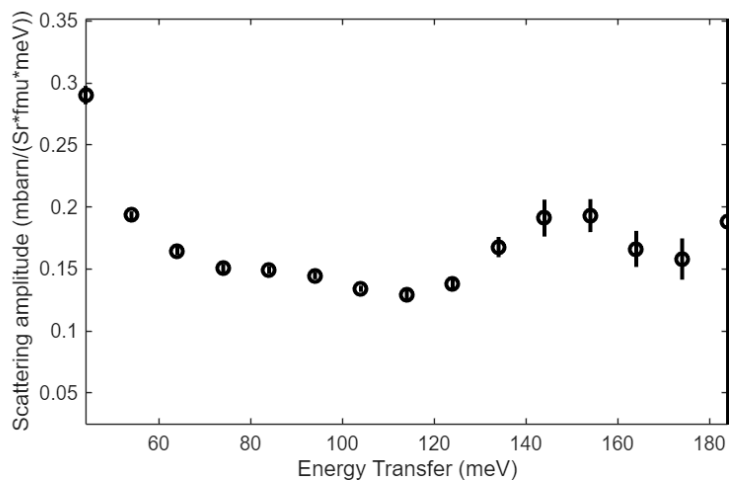
Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.19\pm0.16$; $J_0=3.2\pm0.066$; $\gamma=3.49691\pm3.54998$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $179 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

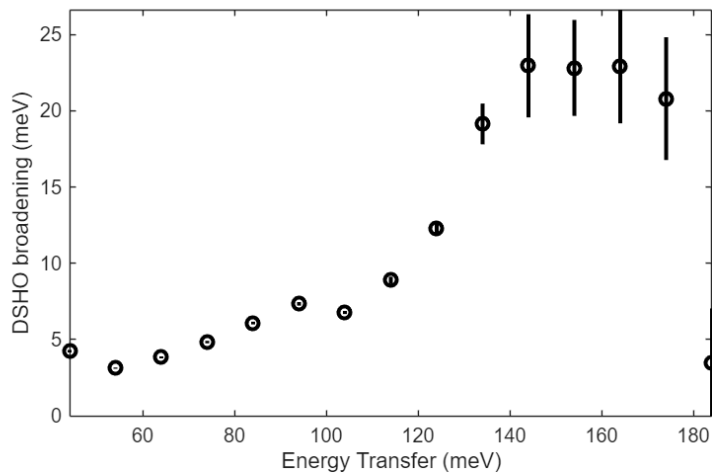
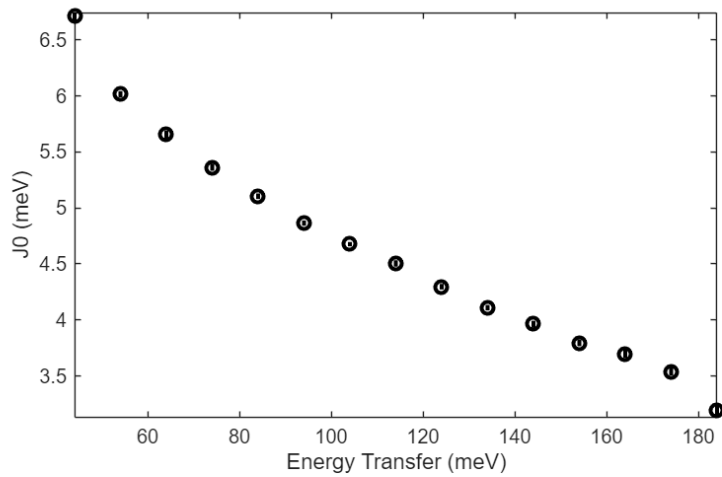


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $179 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



 **** En = 194±5 going to 194





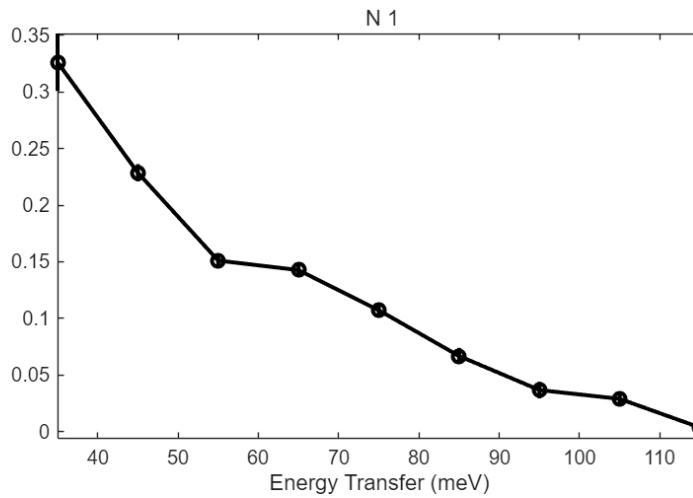
```

c1 = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [c1{:}];

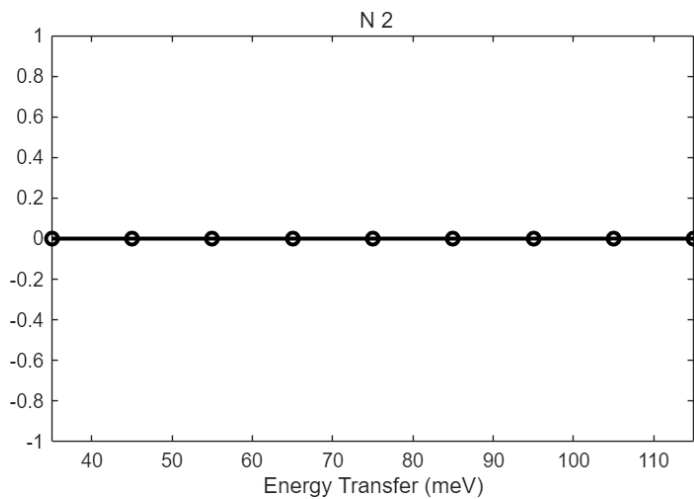
beep
beep
beep

all_fit_par = GP_FFCutsFits.Ei200_off110GPmiff_all_fit_par;
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GP_FFCutsFits;
save(Cutfitfile,'fit_src_struct','-v7.3')
```